

Questions with Answer Keys

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Q1 - Tangent and Normal

The equation of the normal to the circle $x^2 + y^2 - 8x - 2y + 12 = 0$ at the points whose ordinate is -1 , will be

(1) $2x - y - 7 = 0, 2x + y - 9 = 0$

(2) $2x + y - 7 = 0, 2x + y + 9 = 0$

(3) $2x + y + 7 = 0, 2x + y + 9 = 0$

(4) $2x - y + 7 = 0, 2x - y + 9 = 0$

Q2 - Tangent and Normal

The value of c for which $x + y = 2$ touches circle $x^2 + y^2 - 2cx = 0$ and $(1,0)$ lies inside the circle, is

(1) 2

(2) $\sqrt{2} - 1$

(3) $2(\sqrt{2} - 1)$

(4) $2(\sqrt{2} + 1)$

Q3 - Tangent and Normal

Tangents drawn from the point $P(1, 8)$ to the circle $x^2 + y^2 - 6x - 4y - 11 = 0$ touch the circle at the points A and B . The equation of the circumcircle of the ΔPAB , is

(1) $x^2 + y^2 + 4x - 6y + 19 = 0$

(2) $x^2 + y^2 - 4x - 10y + 19 = 0$

(3) $x^2 + y^2 - 2x + 6y - 29 = 0$

(4) $x^2 + y^2 - 6x - 4y + 19 = 0$

Q4 - Tangent and Normal

Let C be the circle with centre $(0,0)$ and radius 3 units. The equation of the locus of the mid-points of the chords of the circle C that subtends an angle of $\frac{2\pi}{3}$ at its centre, is

(1) $x^2 + y^2 = 1$

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(2) $x^2 + y^2 = \frac{27}{4}$

(3) $x^2 + y^2 = \frac{9}{4}$

(4) $x^2 + y^2 = \frac{3}{2}$

Q5 - Tangent and Normal

Two parallel chords of a circle of radius 2 are at a distance $(\sqrt{3} + 1)$ apart. If the chords subtends at the centre, angles of $\frac{\pi}{k}$ and $\frac{2\pi}{k}$, where $k > 0$, then the value of $[k]$, is (where $[.]$ denotes Greatest Integer Function)

(1) 2

(2) 3

(3) 1

(4) 0

Q6 - Angle Between Circles and Radical Axis

A point circle has its centre at (2,-2). The radical axis of this circle with the circle

$$3(x^2 + y^2) - 6x + 9y + 1 = 0, \text{ is}$$

(1) $6x - 3y + 23 = 0$

(2) $6x + 3y - 23 = 0$

(3) $6x + 3y + 23 = 0$

(4) $6x - 3y - 23 = 0$

Q7 - Angle Between Circles and Radical Axis

The radical centre of circles $(x - p)^2 + y^2 = p^2$, $x^2 + (y - q)^2 = q^2$ and $x^2 + y^2 = p^2 + q^2$, is

(1) $\left(\frac{p^2+q^2}{2q}, \frac{p^2+q^2}{2p}\right)$

(2) $\left(\frac{p^2+q^2}{p}, \frac{p^2+q^2}{q}\right)$

(3) $\left(\frac{p^2+q^2}{2p}, \frac{p^2+q^2}{2q}\right)$

(4) None of these

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Q8 - Angle Between Circles and Radical Axis

Two equal circles of radius r , intersect such that each passes through the centre of the other. The length of their common chord is

(1) $\sqrt{2}r$

(2) $2r$

(3) $\sqrt{3}r$

(4) $\frac{\sqrt{3}r}{2}$

Q9 - Angle Between Circles and Radical Axis

The radical centre of the three circles $x^2 + y^2 = a^2$, $(x - c)^2 + y^2 = a^2$ and $x^2 + (y - b)^2 = a^2$, is

(1) $\left(\frac{a}{2}, \frac{b}{2}\right)$

(2) $\left(\frac{b}{2}, \frac{c}{2}\right)$

(3) $\left(\frac{c}{2}, \frac{b}{2}\right)$

(4) None of these

Q10 - Angle Between Circles and Radical Axis

The equation of radical axis of the circle $x^2 + y^2 + x - y + 2 = 0$ and $3x^2 + 3y^2 - 4x - 12 = 0$ will be

(1) $2x^2 + 2y^2 - 5x + y - 14 = 0$

(2) $7x - 3y + 18 = 0$

(3) $5x - y + 14 = 0$

(4) None of these

Q11 - Angle Between Circles and Radical Axis

If the circles of same radius r and centres at $(3,4)$ and $(7,8)$ cut orthogonally, then r is (in units)

(1) 2

(2) 3

(3) 4

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(4) 5

Q12 - Angle Between Circles and Radical Axis

If a circle passes through the point (1,2) and cuts the circle $x^2 + y^2 = 4$ orthogonally, then the equation of the locus of its centre is

(1) $x^2 + y^2 - 2x - 4y - 7 = 0$

(2) $x^2 + y^2 - 3x - 8y + 1 = 0$

(3) $2x + 4y - 9 = 0$

(4) $2x + 4y - 1 = 0$

Q13 - Location of Circles

The number of common tangents to the circles $x^2 + y^2 = 4$ and $x^2 + y^2 - 6x - 8y = 24$, is

(1) 0

(2) 1

(3) 3

(4) 4

Q14 - Location of Circles

Two circles $x^2 + y^2 + 2hx = r_1^2 - h^2$ and $x^2 + y^2 + 2kx = r_2^2 - k^2$ touch externally, if

(1) $h^2 + k^2 = r_1^2 - r_2^2$

(2) $h^2 + k^2 = r_1^2 + r_2^2$

(3) $h^2 + k^2 = r_1^2 + r_2^2 - 2r_1r_2$

(4) $h^2 + k^2 = r_1^2 + r_2^2 + 2r_1r_2$

Q15 - Location of Circles

If a circle S_1 bisects the circumference of another circle S_2 , then their radical axis

(1) passes through the centre of S_1

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(2) passes through the centre of S_2

(3) bisects the line joining their centres

(4) None of the above

Q16 - Location of Circles

The locus of a point such that tangents drawn from it to the circle $x^2 + y^2 = a^2$ are at 120° , is

(1) $4x^2 + 4y^2 = 3a^2$

(2) $3x^2 + 3y^2 = 4a^2$

(3) $x^2 + y^2 = \frac{2a^2}{3}$

(4) None of these

Q17 - Location of Circles

If the circle $C_1 : x^2 + y^2 = 16$ intersects another circle of C_2 of radius 5 in such a manner that the common chord is of maximum length and has a slope equal to $\frac{3}{4}$, then the coordinates of the centre of C_2 , are

(1) $\left(\frac{-9}{5}, \frac{-12}{5}\right)$

(2) $\left(\frac{9}{5}, \frac{12}{5}\right)$

(3) $\left(\pm \frac{9}{5}, \pm \frac{12}{5}\right)$

(4) None of these

Q18 - Location of Circles

Let $ABCD$ be a quadrilateral with area 18, with side AB parallel to the side CD and $AB = 2CD$. Let AD be the perpendicular to AB and CD . If a circle is drawn inside the quadrilateral $ABCD$ touching all the sides, then its radius is (in units)

(1) 3

(2) 2

(3) $\frac{3}{2}$

(4) 1

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Q19 - Chord of Contact, Chord with Given Mid Point & Pair of tangents

Equation of pair of tangents from $(0,1)$ on circle $x^2 + y^2 = \frac{1}{4}$, is

(1) $\left(x^2 + y^2 - \frac{1}{4}\right) \cdot \left(\frac{3}{4}\right) = \left(y - \frac{1}{4}\right)^2$

(2) $\left(x^2 - y^2 - \frac{1}{4}\right) \cdot \left(\frac{3}{4}\right) = \left(y - \frac{1}{4}\right)^2$

(3) $\left(x^2 + y^2 - \frac{1}{4}\right) \cdot \left(\frac{3}{4}\right) = \left(y + \frac{1}{4}\right)^2$

(4) None of these

Q20 - Chord of Contact, Chord with Given Mid Point & Pair of tangents

If two distinct chords from the point (p, q) on the circle $x^2 + y^2 = px + qy$ (where, $pq \neq 0$) are bisected by the X -axis, then

(1) $p^2 = q^2$

(2) $p^2 = 8q^2$

(3) $p^2 < 8q^2$

(4) $p^2 > 8q^2$

Q21 - Chord of Contact, Chord with Given Mid Point & Pair of tangents

Tangents are drawn from the point on the circle $x^2 + y^2 = a^2$ to the circle $x^2 + y^2 = b^2$ and the chord of contact touch the circle $x^2 + y^2 = c^2$, then a, b, c will be in

(1) AP

(2) GP

(3) HP

(4) None of these

Q22 - Chord of Contact, Chord with Given Mid Point & Pair of tangents

The locus of the mid-point of the chord of tangents drawn from points lying on the straight line $4x - 5y = 20$ to the circle $x^2 + y^2 = 9$, is

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(1) $20(x^2 + y^2) - 36x + 45y = 0$

(2) $20(x^2 + y^2) + 36x - 45y = 0$

(3) $36(x^2 + y^2) - 20x + 45y = 0$

(4) $36(x^2 + y^2) + 20x - 45y = 0$

Q23 - Chord of Contact, Chord with Given Mid Point & Pair of tangents

The equation of the pair of tangents drawn to the circle $x^2 + y^2 - 2x + 4y + 3 = 0$ from (6,-5) is

(1) $7x^2 + 23y^2 + 30xy + 66x + 50y - 73 = 0$

(2) $7x^2 + 23y^2 - 30xy - 66x - 50y + 73 = 0$

(3) $7x^2 + 23y^2 + 30xy - 66x - 50y - 73 = 0$

(4) None of the above

Q24 - Chord of Contact, Chord with Given Mid Point & Pair of tangents

The equation of the locus of the middle point of a chord of the circle $x^2 + y^2 - 2x - 2y = 0$ such that the pair of lines joining the origin to the point of intersection of the chord and the circle are equally inclined to the X - axis is

(1) $x + y = 2$

(2) $x - y = 2$

(3) $2x - y = 1$

(4) None of these

Q25 - Chord of Contact, Chord with Given Mid Point & Pair of tangents

Three concentric circles of which the biggest is $x^2 + y^2 = 1$, have their radii in an AP. If the line $y = x + 1$ cuts all the circles in real and distinct points. The interval in which the common difference of the AP will lie is

(1) $\left(0, \frac{1}{4}\right)$

(2) $\left(0, \frac{1}{2\sqrt{2}}\right)$

(3) $\left(0, \frac{2-\sqrt{2}}{4}\right)$

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(4) None of these

Q26 - Pole and Polar

The pole of the line $9x + y - 28 = 0$ w.r.t. circle $2x^2 + 2y^2 - 3x + 5y - 7 = 0$, is

(1) (3,1)

(2) (1,3)

(3) (3,-1)

(4) (-3,1)

Q27 - Pole and Polar

If line $x + 2y = 1$ is the polar of circle $x^2 + y^2 = 5$, then pole is

(1) (5,10)

(2) (5,5)

(3) (-5,10)

(4) (-5,-10)

Q28 - Pole and Polar

If the pole of the polar w.r.t. the circle $x^2 + y^2 = c^2$ is on the circle $x^2 + y^2 = 9c^2$. Then, this polar will be the tangent of the circle.

(1) $x^2 + y^2 = \frac{9}{4}c^2$ (2) $x^2 + y^2 = \frac{c^2}{9}$ (3) $x^2 + y^2 = 4c^2$

(4) None of these

Q29 - Pole and Polar

The lengths of the tangents from the points A and B to a circle are l_1 and l_2 , respectively. If points are conjugate with respect to the circle, then AB^2 is equal to

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(1) $l_1 + l_2$

(2) $l_1^2 + l_2^2$

(3) $|l_1^2 - l_2^2|$

(4) None of these

Q30 - Pole and Polar

The locus of poles whose polar with respect to $x^2 + y^2 = a^2$ always passes through $(k, 0)$, is

(1) $kx = a^2$

(2) $kx + a^2 = 0$

(3) $ky + a^2 = 0$

(4) $ky = a^2$

